

The Gauntlet — Instructor Guide

Forensics CLI 101 · every challenge, answer, and accepted variant · generated 2026-08-17 from the live game data

Quick reference

Site: `gauntlet-cis4400.netlify.app` · Class password: `locard-2026` (lowercase, case-sensitive)
Handles: 3–20 characters, letters/numbers/underscore/hyphen, **claimed once only**. Admin handle: `warden` .
30 challenges · 885 points total · flags look like `FLAG{ABCD2345EFGH}` and are **different for every student**.

If a student says "it's not working"

Symptom	Cause & what to say
"My flag is rejected"	Flags are per-student — a classmate's flag never validates. Tell them to click the green flag in the terminal to copy it exactly, rather than retyping.
"Handle already claimed"	Someone took it, or they registered before and cleared their browser. Pick a new handle. (To free one, delete that player in Netlify Blobs.)
"I ran the command, nothing happened"	Check the blue path in their prompt — they may be in a different directory. Most challenges accept short or full paths, but the file must exist from where they stand. <code>cd ~</code> returns home.
"It says command not found"	If it's a real Linux command we don't simulate (<code>top</code> , <code>nano</code>), the game says so explicitly. Genuine typos say "command not found".
"The act is locked"	Each act opens after solving all but one of the previous act's challenges. Topside (WIN) is always open.
"Nothing printed"	Correct for <code>cd</code> and redirects. The game prints "(no output — in a shell, silence usually means success)".
Lost all progress	Progress is tied to the browser token. Same browser = resumes automatically. Different machine = new handle needed.

Act structure & unlocking

Act	Teaches	Challenges	Points	Unlock
Act I: First Steps	Basic navigation, paths, prompt, and directory structure	7	90	Always open
Act II: Reading the Evidence	Inspecting files, headers, and forensic checksums	6	120	Solve 6 of 7 in the prior act
Act III: Search & Discovery	Pattern search, find, manual pages, and the WSL /mnt/c bridge	7	235	Solve 5 of 6 in the prior act
Act IV: The Plumbing	Pipes, redirection, filters, and multi-stage analysis	4	180	Solve 6 of 7 in the prior act
Act V: The Capstone	Multi-step investigation: partition offsets and evidence carving	2	180	Solve 3 of 4 in the prior act
Topside (Windows Quest)	Windows CMD parity: <code>dir /a</code> , <code>type</code> , <code>findstr</code> , <code>attrib</code> , and <code>certutil</code>	4	80	Always open

Students may skip **one** challenge per act and still advance. The Topside (Windows) quest is optional and open from the start.

Act I: First Steps

Basic navigation, paths, prompt, and directory structure

Where Am I?

10 pts · act1-pwd · OK

Task: Use the `pwd` command to print the directory you are standing in.

Answer: `pwd`

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Type `pwd` (Present Working Directory) and press Enter.

Survey the Ground

10 pts · act1-ls · OK

Task: Use `ls` to list the files and directories where you are standing.

Answer: `ls`

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Type `ls` to list files and folders.

Hidden in Plain Sight

15 pts · act1-hidden · OK

Task: Something is hiding in your home directory. In Linux, files beginning with `.` are invisible to a plain `ls`. Use `ls -la` to find `.stash`, then read it with `cat .stash`. To submit the flag: click the `FLAG{...}` in the terminal to copy it, then type `submit` and paste it — or use the box in the left panel.

Answer: `ls -la`

Then: `cat .stash`

Completion: Submits a flag · flag is inside `/home/analyst/.stash`

Hint 1 (free): Run `ls -la` or `ls -a` to reveal files starting with a dot (`.`).

Hint 2 (-5 XP): Run `cat .stash` to read the hidden file, then copy the `FLAG{...}` value and run `submit FLAG{...}`.

One Level Down

15 pts · act1-cd · OK

Task: Move into `training/level_1`. Read `checkpoint_alpha.txt` and submit the recovered flag.

Answer: `cd training/level_1`

Then: `cat checkpoint_alpha.txt`

Completion: Submits a flag · flag is inside `/home/analyst/training/level_1/checkpoint_alpha.txt`

Hint 1 (free): Use `cd training/level_1` to move down into it.

Hint 2 (-5 XP): Once you are inside `level_1`, run `cat checkpoint_alpha.txt` to view the flag, then run `submit FLAG{...}`. (From home, the full path works too: `cat training/level_1/checkpoint_alpha.txt`.)

Sideways Move

15 pts · act1-paths · OK

Task: Now reach `training/level_2` — a SIBLING of `level_1`, not a child. If you are inside `level_1`, you must go up before you can go over. Read `checkpoint_beta.txt` and submit the flag.

Answer: `cd training/level_2`

Then: `cat checkpoint_beta.txt`

Completion: Submits a flag · flag is inside `/home/analyst/training/level_2/checkpoint_beta.txt`

Hint 1 (free): If you are inside `level_1`, run `cd ..` to go up to `training`, then `cd level_2`. From home, `cd training/level_2` gets you there in one step.

Hint 2 (-5 XP): Once you are inside `level_2`, run `cat checkpoint_beta.txt` and submit the flag it contains. (From home: `cat training/level_2/checkpoint_beta.txt`.)

Instructor note: From inside level_1 the sibling route is: `cd ..` then `cd level_2` — that is the lesson.

Warp Speed (Tab Completion)

15 pts · act1-tab · OK

Task: Typing full directory paths is slow. Type `cd Doc` and press Tab to auto-complete `cd Documents`, then press Enter.

Answer: `cd Doc`

Then: press Tab → completes to Documents, then Enter

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Type `cd Doc` then press the Tab key. The shell will auto-complete to `cd Documents`.

Instructor note: Students type `cd Doc` and press Tab. Typing it in full also counts.

Recall the Past (History)

10 pts · act1-history · OK

Task: Your recent commands are saved. Press the Up Arrow key to bring back a command you ran earlier — like `pwd` — then press Enter to run it without retyping. (Running `pwd` completes this challenge; the Up-Arrow habit is the real lesson.)

Answer: `pwd`

Then: press Up Arrow, then Enter

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Run `pwd`, then press Up Arrow and Enter.

Act II: Reading the Evidence

Inspecting files, headers, and forensic checksums

Read the Case File

15 pts · act2-cat · OK

Task: Every investigation starts by opening the case file. If your prompt shows you are not in `/home/analyst`, run `cd ~` first. Then use `cat` to display `Documents/case_notes.txt` on screen. Reading it is the whole task — the challenge completes the moment you do.

Answer: `cat Documents/case_notes.txt`

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Type `cat Documents/case_notes.txt` — that single command is the entire challenge.

The Earliest Anomaly

15 pts · act2-head · OK

Task: `Documents/access.log` contains extensive records. Use `head -n 5 Documents/access.log` to view only the first 5 log entries.

Answer: `head -n 5 Documents/access.log`

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Run: `head -n 5 Documents/access.log`

The Sensor Freeze

20 pts · act2-tail · OK

Task: The critical incident occurred right before sensors went dark. Use `tail` on `Documents/access.log` to read the final log entries and submit the flag.

Answer: `tail Documents/access.log`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/access.log`

Hint 1 (free): Run `tail -n 10 Documents/access.log` or `tail Documents/access.log`.

Hint 2 (-5 XP): Find the flag on the final line and run `submit FLAG{...}`.

Don't Trust Extensions

20 pts · act2-file · OK

Task: Never trust a filename — analysts identify files by their contents. Use `file evidence/mystery_file` to reveal its true format. (`file` reads the first few bytes of a file — the "magic bytes" — which identify it no matter what it is named.)

Answer: `file evidence/mystery_file`

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Run `file evidence/mystery_file`

Carving Binary Artifacts

25 pts · act2-strings · OK

Task: `evidence/binary_data` is an executable containing compiled instructions. Use `strings evidence/binary_data` to extract embedded readable text and submit the discovered flag.

Answer: `strings evidence/binary_data`

Completion: Submits a flag · flag is inside `/home/analyst/evidence/binary_data`

Hint 1 (free): Run `strings evidence/binary_data`

Hint 2 (-5 XP): Look for `TARGET_FLAG=FLAG{...}` in the output and submit it.

Integrity Checksum (MD5)

25 pts · act2-md5 · OK

Task: Digital evidence must be hashed to prove it was not altered. Step 1: run `md5sum evidence/evidence.img` to compute its hash. The hash is NOT the flag. Step 2: run `cat evidence/evidence.img` to read the image header, find the `FLAG{...}` inside, and submit it.

Answer: `md5sum evidence/evidence.img`

Then: `cat evidence/evidence.img`

Completion: Submits a flag · flag is inside `/home/analyst/evidence/evidence.img`

Hint 1 (free): Run `md5sum evidence/evidence.img` to compute the hash.

Hint 2 (-5 XP): Run `cat evidence/evidence.img`. The `FLAG{...}` is on the "Payload signature" line — click it to copy, then run `submit FLAG{...}`.

Instructor note: The hash is NOT the flag — the flag is in the file text.

Act III: Search & Discovery

Pattern search, find, manual pages, and the WSL /mnt/c bridge

Secrets in Plain Sight

25 pts · act3-grep · OK

Task: Search `Documents/secrets.txt` for the line containing `vault_passcode` using `grep`, then submit the flag.

Answer: `grep vault_passcode Documents/secrets.txt`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/secrets.txt`

Hint 1 (free): Run: `grep vault_passcode Documents/secrets.txt`

Hint 2 (-5 XP): Submit the resulting `FLAG{...}` flag.

Case-Insensitive Searching

30 pts · act3-grep · OK

Task: `Documents/logs.txt` logs errors inconsistently as "error", "ERROR", and "Error". Use `grep -i "error" Documents/logs.txt` to see every error line regardless of capitalization, then submit the `FLAG{...}` you find on one of them.

Answer: `grep -i "error" Documents/logs.txt`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/logs.txt`

Hint 1 (free): Use the `-i` flag: `grep -i "error" Documents/logs.txt`

Hint 2 (-5 XP): One of the matching ERROR lines ends with a `FLAG{...}`. Click the flag to copy it, then run `submit FLAG{...}`.

Needle in /var/log

30 pts · act3-find · OK

Task: Locate all log files in `/var/log` using `find /var/log -name "*.log"`. Read the sensor audit log and submit the keycode flag.

Answer: `find /var/log -name "*.log"`

Then: `cat /var/log/sensor_audit.log`

Completion: Submits a flag · flag is inside `/var/log/sensor_audit.log`

Hint 1 (free): Run `find /var/log -name "*.log"` to see every log that exists there.

Hint 2 (-10 XP): Run `cat /var/log/sensor_audit.log` (the full path from the find output), then submit the keycode `FLAG{...}` it contains.

The WSL Bridge

40 pts · act3-crossing · OK

Task: This machine has a Windows side. Its folder `C:\Users\analyst\Desktop\CASE_FILES` is reachable from Linux because WSL mounts drive C: at `/mnt/c`. Read `/mnt/c/Users/analyst/Desktop/CASE_FILES/intake.txt` and submit the flag.

Answer: `cat /mnt/c/Users/analyst/Desktop/CASE_FILES/intake.txt`

Completion: Submits a flag · flag is inside `/mnt/c/Users/analyst/Desktop/CASE_FILES/intake.txt`

Hint 1 (free): WSL mounts Windows drives under `/mnt/c`. Check `ls /mnt/c/Users/analyst/Desktop/CASE_FILES`.

Hint 2 (–10 XP): Run `cat /mnt/c/Users/analyst/Desktop/CASE_FILES/intake.txt` and submit the verification flag.

The Bridge, Unassisted

45 pts · act3-crossing-solo · OK

Task: A second dossier sits on the Windows side at `C:\Users\analyst\Documents\surface_notes.txt`. Reach it from this Linux shell and submit the flag inside. This time, no path translation is given — Case 001 will not give you one either.

Answer: `cat /mnt/c/Users/analyst/Documents/surface_notes.txt`

Completion: Submits a flag · flag is inside `/mnt/c/Users/analyst/Documents/surface_notes.txt`

Hint 1 (free): Windows drives are mounted somewhere under the Linux root. Start with `ls /` and explore.

Hint 2 (–15 XP): `ls /mnt` shows the mounted drives. Translate the rest of the Windows path into Linux form yourself: `C:\` becomes `/mnt/c/`, and backslashes become forward slashes.

Instructor note: Deliberately unassisted — mirrors Case 001 §1C.

Consult the Manual

30 pts · act3-man · OK

Task: Real forensic analysts read the manual. Use `man tracker` to inspect the on-line manual page for the tracker sensor suite, uncover the secret manual override code, and submit it.

Answer: `man tracker`

Completion: Submits a flag

Hint 1 (free): Run `man tracker` in the terminal.

Hint 2 (–10 XP): Read the DESCRIPTION section carefully — the override code is buried in it. Submit the `FLAG{...}` code you find there.

Instructor note: Flag is inside the DESCRIPTION section.

Deploy the Sensor Suite (apt-get)

35 pts · act3-apt · OK

Task: The `tracker` tool is not yet installed on this node. Run `sudo apt-get update && sudo apt-get install tracker -y` to install it, then execute `tracker -a` to run a full sensor sweep and submit the flag.

Answer: `sudo apt-get update && sudo apt-get install tracker -y`

Then: `tracker -a`

Completion: Submits a flag · flag is inside `tracker`

Hint 1 (free): Run: `sudo apt-get update && sudo apt-get install tracker -y`

Hint 2 (–10 XP): Once installed, run `tracker -a` and submit the returned flag.

Act IV: The Plumbing

Pipes, redirection, filters, and multi-stage analysis

Filter the Noise (Inverted Grep)

40 pts · act4-grep-v · OK

Task: Start from home (`cd ~` if needed). `Documents/network_stream.log` is filled with normal `ALLOW` traffic. Use `grep -v "ALLOW" Documents/network_stream.log` to filter out benign packets, isolate the critical leak packet, and submit the flag.

Answer: `grep -v "ALLOW" Documents/network_stream.log`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/network_stream.log`

Hint 1 (free): Run: `grep -v "ALLOW" Documents/network_stream.log`

Hint 2 (–10 XP): Find the `CRITICAL_DATA_LEAK` entry and submit the flag.

First Pipeline (grep | wc -l)

45 pts · act4-pipe-count · OK

Task: Pipelines connect commands together using `|`. Pipe non-allowed network packets into `wc -l` to count how many anomalies exist: `grep -v "ALLOW" Documents/network_stream.log | wc -l`.

Answer: `grep -v "ALLOW" Documents/network_stream.log | wc -l`

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Type: `grep -v "ALLOW" Documents/network_stream.log | wc -l`

Extracting Fields (cut & grep)

45 pts · act4-pipe-csv · OK

Task: `Documents/security_events.csv` holds comma-delimited logs. Pipe `grep "FLAG_EMIT" Documents/security_events.csv | cut -d, -f6` to extract the flag column and submit the token.

Answer: `grep "FLAG_EMIT" Documents/security_events.csv | cut -d, -f6`

Completion: Submits a flag · flag is inside `/home/analyst/Documents/security_events.csv`

Hint 1 (free): Run `grep "FLAG_EMIT" Documents/security_events.csv | cut -d, -f6` or `cat Documents/security_events.csv | grep FLAG_EMIT`.

Hint 2 (–10 XP): Submit the extracted `FLAG{...}` flag.

Output Redirection (>)

50 pts · act4-redirect · OK

Task: The `>` operator redirects output to a file on disk instead of the screen. Extract all ERROR lines from `Documents/logs.txt` and save them into `/tmp/errors.log`: `grep -i "error" Documents/logs.txt > /tmp/errors.log`. Then `cat /tmp/errors.log`.

Answer: `grep -i "error" Documents/logs.txt > /tmp/errors.log`

Then: `cat /tmp/errors.log`

Completion: Auto-completes when the file is created (no flag)

Hint 1 (free): Run: `grep -i "error" Documents/logs.txt > /tmp/errors.log`

Hint 2 (–10 XP): Run the redirect command FIRST (it prints nothing — that is correct). Then run `cat /tmp/errors.log` to see what got saved into the file.

Instructor note: The redirect prints nothing — that is correct.

Act V: The Capstone

Multi-step investigation: partition offsets and evidence carving

Partition Geometry Inspection

60 pts · act5-scan · OK

Task: Raw forensic image `evidence/suspect_drive.raw` contains multiple partition slices. Run `scan evidence/suspect_drive.raw` to analyze the partition table and identify the starting sector offset of the Evidence Vault partition. (A sector offset is the sector number where a partition begins — a bookmark inside the disk image.)

Answer: `scan evidence/suspect_drive.raw`

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Run: `scan evidence/suspect_drive.raw`

Instructor note: Students must READ the Start sector (206848) from the table.

Carve the Evidence (Finish Line)

120 pts · act5-capstone · OK

Task: Take the sector offset `206848` discovered in the partition scan and pass it to the forensic carver: `extract -o 206848 evidence/suspect_drive.raw`. Submit the decrypted Master Capstone Flag.

Answer: `extract -o 206848 evidence/suspect_drive.raw`

Completion: Submits a flag · flag is inside `act5-capstone`

Hint 1 (free): Run: `extract -o 206848 evidence/suspect_drive.raw`

Hint 2 (–20 XP): Submit the master flag `FLAG{...}` printed in the terminal.

Topside (Windows Quest)

Windows CMD parity: `dir`, `/a`, `type`, `findstr`, `attrib`, and `certutil`

Windows Navigation (`dir` & `cd`)

15 pts · topside-nav · OK

Task: In Windows CMD, `cd` alone displays your current working directory, and `dir` lists files. Run `dir` to inspect `C:\Users\Analyst`.

Answer: `dir`

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Type `dir` and press Enter.

Hidden File Attributes (`attrib`)

20 pts · topside-attrb · OK

Task: Windows uses attributes rather than dotfiles for hidden files. Run `attrib evidence\mystery_file`, then `type evidence\mystery_file` to recover the hidden Topside flag.

Answer: `attrib evidence\mystery_file`

Then: `type evidence\mystery_file`

Completion: Submits a flag · flag is inside `C:\Users\Analyst\evidence\mystery_file`

Hint 1 (free): Run `attrib evidence\mystery_file` to view its H (Hidden) attribute.

Hint 2 (-5 XP): Run `type evidence\mystery_file` and submit the flag.

Windows String Search (`findstr /i`)

25 pts · topside-findstr · OK

Task: Use `findstr /i "marker" Documents\logs.txt` to search the Windows event log for the flagged event and submit the flag.

Answer: `findstr /i "marker" Documents\logs.txt`

Completion: Submits a flag · flag is inside `C:\Users\Analyst\Documents\logs.txt`

Hint 1 (free): Run: `findstr /i "marker" Documents\logs.txt`

Hint 2 (-5 XP): Submit the discovered flag.

CertUtil Hashing

20 pts · topside-certutil · OK

Task: In Windows environments, `certutil -hashfile <file> MD5` is used for forensic integrity checks. Calculate the MD5 hash of `evidence\evidence.img`.

Answer: `certutil -hashfile evidence\evidence.img MD5`

Completion: Auto-completes on the command (no flag)

Hint 1 (free): Run: `certutil -hashfile evidence\evidence.img MD5`

Accepted command variants

Auto-completing challenges accept the natural variations a beginner types, so a correct command is never silently ignored:

- Optional quotes: `cat "Documents/case_notes.txt"`

- Path prefixes: `./Documents` , `~/Documents` , `/home/analyst/Documents`
- Short paths when already in the folder: `head -n 5 access.log` from inside `Documents`
- Flag spacing: `head -n 5` , `head -n5` , `head -5`
- Windows: forward or back slashes, any capitalization (`md5` or `MD5`)
- Trailing spaces are ignored everywhere

Teaching notes

- **The Coach** (toggle in the terminal title bar) explains every command and error in plain language. Explanations fade after the second use of a command. Leave it ON for beginners.
- **Reference tab** holds a manual page for every command, and tells students the real-shell equivalent (`man grep`). Point stuck students there rather than giving the answer.
- **Course tie-ins:** Act III's WSL bridge is exactly what Case 001 §1C asks students to figure out unaided. Act V's offset-carrying mirrors Case 003's `mmls` → `fsstat -o` chain. Act IV's pipes and redirection are Case 005's core skills.
- **Unsimulated tools** (`mmls` , `vol` , `exiftool`) reply with what they are and which case uses them — a free preview of the semester.
- **Hints:** first is free, later ones cost XP. Hint use is visible in the instructor dashboard as a signal of where the class is struggling.